

F O R T I M A N A G E R N S E 6

Lab 01 — Deployment and Initial Configuration

01. Deployment and Management Access

A FortiManager arrives with no useful network identity. Before anything else can happen it needs a reachable management address, and the first decision — how that address is obtained — carries further than it appears.

► 1.1 Default access

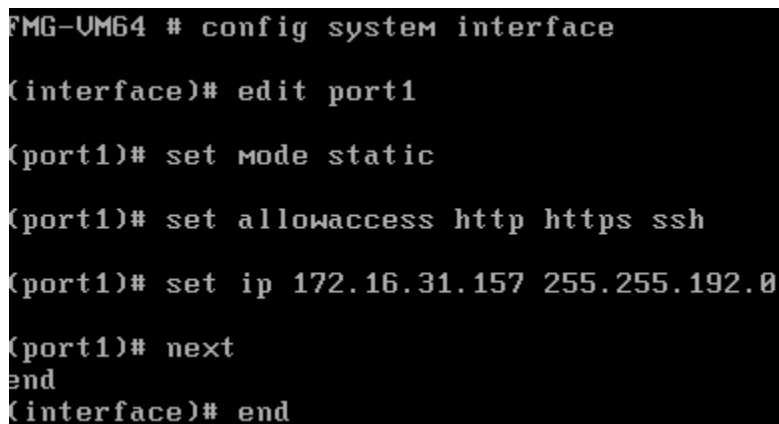
A hardware FortiManager ships with its management port reachable at 192.168.1.99. A VM deployment has no address at all until one is assigned, so the first session is always on the console rather than over the network. On this platform the management interface is port1.

► 1.2 Assigning the address

Two options exist, and the choice matters more on a management appliance than on an ordinary host.

DHCP is quick and is what this lab used initially. It is the wrong answer in production: every managed FortiGate is configured to reach FortiManager at a fixed address, so a lease change silently breaks every management tunnel at once. A static address is the correct deployment practice.

```
config system interface
  edit port1
    set mode static
    set ip 172.16.31.157 255.255.192.0
    set allowaccess https ssh http
  next
end
```



```
FMG-VM64 # config system interface
(interface)# edit port1
(port1)# set mode static
(port1)# set allowaccess http https ssh
(port1)# set ip 172.16.31.157 255.255.192.0
(port1)# next
end
(interface)# end
```

Figure 1 — Assigning a static management address to port1 from the console.

```
FMG-VM64 # get system interface port1
name                : port1
status              : enable
mode                : static
ip                  : 172.16.31.183 255.255.192.0
allowaccess         : https ssh http
serviceaccess      :
lldp                : disable
speed               : auto
description         : (null)
alias               : (null)
mtu                 : 1500
type                : physical
ipv6:
  ip6-address: ::1/0
  ip6-allowaccess:
  ip6-autoconf: enable
```

Figure 2 — Verifying the interface state with `get system interface port1`.

A note on allowaccess

Plain `http` is enabled above so that the lab can be reached without certificate warnings. It should not be enabled on a production management appliance: the FortiManager admin session carries credentials with authority over every managed firewall, and in cleartext those credentials cross the network in the clear.

The production form is set `allowaccess https ssh`, with HTTP-to-HTTPS redirection enabled in the admin settings.

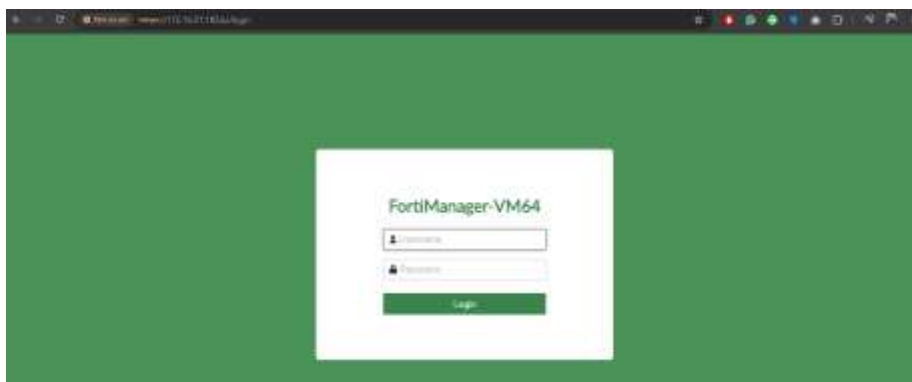


Figure 3 — The FortiManager login page, reachable once the interface is configured.

02. Licensing

FortiManager-VM will not manage a single device until it is licensed. The appliance boots, the GUI loads, and the login page offers activation rather than access.

18:42

 https://10.58.228.22/ui/

+

33



FortiManager-VM64

This product requires a valid license.
You could log in to FortiCloud to
activate your purchased license, or use
a free trial license.

 Account ID/Email

 Password

☒ Free Trial

☐ Activate License

Login with FortiCloud

OR

Register with FortiCloud

Upload License

Figure 4 — The pre-licence login screen, offering FortiCloud activation or a free trial.

Activation requires outbound internet access to FortiCare. This lab runs on an isolated network, so activation was performed by temporarily tethering to a mobile connection, after which the appliance returned to the isolated segment. The licence is validated at activation and re-checked periodically, so an appliance that is permanently offline needs a different approach — covered in section 6.

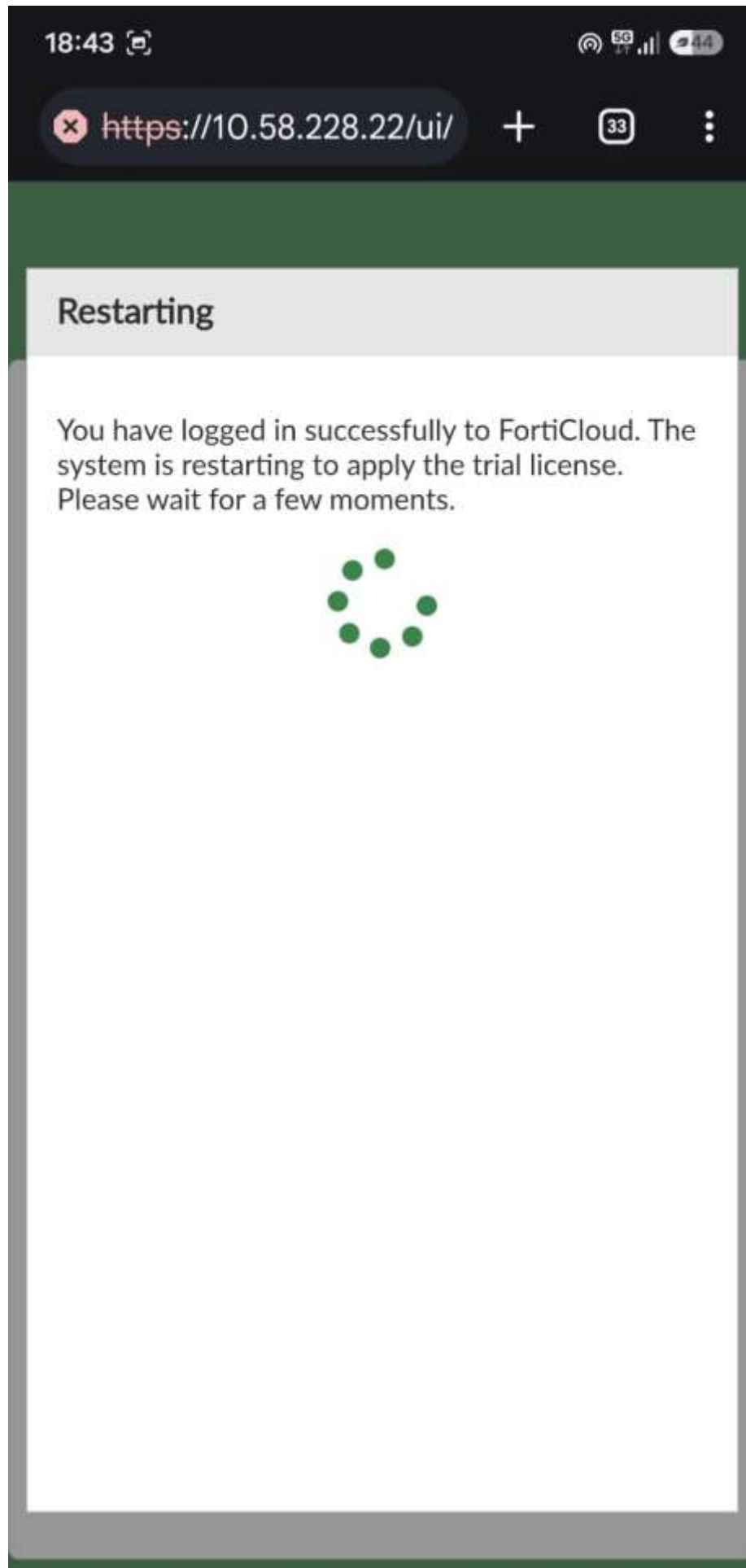


Figure 5 — Completing registration through the FortiCloud portal.

```

Password:
FMG-UM64 # get system status
Platform Type           : FMG-UM64
Platform Full Name      : FortiManager-UM64
Version                 : v7.6.2-build3415 241212 (GA.F)
Serial Number           : FMG-UMTM26811676
BIOS version            : 04000002
Hostname                : FMG-UM64
Max Number of Admin Domains : 2
Max Number of Device Groups : 3
Admin Domain Configuration : Disabled
FIPS Mode               : Disabled
HA Mode                 : Stand Alone
Branch Point            : 3415
Release Version Information : GA.F
Current Time             : Sat Oct 03 07:45:03 PDT 2026
Daylight Time Saving     : Yes
Time Zone               : (GMT-8:00) Pacific Time (US & Canada).
x86-64 Applications      : Yes
Disk Usage               : Free 470.97GB, Total 491.08GB
File System              : Ext4
License Status           : Valid
Image Signature          : Image is GA Certified

```

Figure 6 — The FortiCare activation page.

► 2.1 What the trial licence allows

The dashboard reports the limits directly, and they shape what the rest of this lab series can demonstrate.

Resource	This appliance	Consequence for the series
Managed devices	0 / 3	Enough for policy packages, per-device mapping, and installation targets
ADOMs	0 / 2	Enough for ADOM separation; not enough for a global ADOM above two child ADOMs
FortiGuard services	Not licensed	FortiManager cannot act as a local FDS without a service contract
Licence type	Trial	Time limited — the lab series is scoped to complete within the trial window

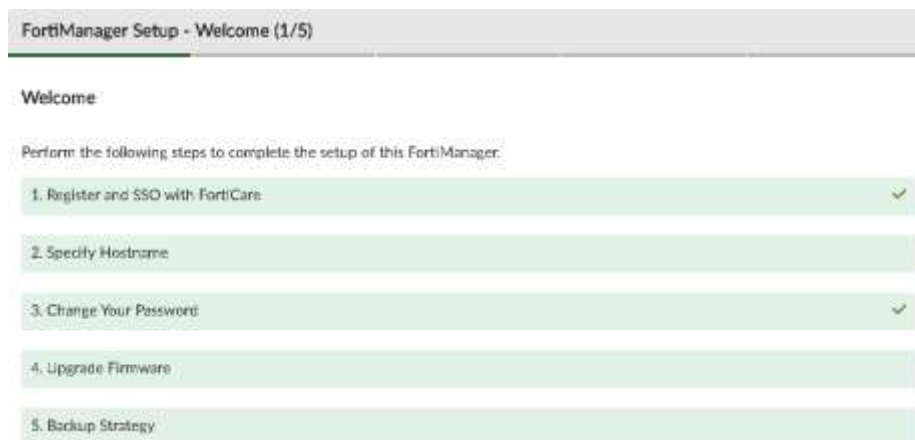


Figure 7 — The post-license setup wizard, which walks through registration, hostname, password, firmware, and backup.

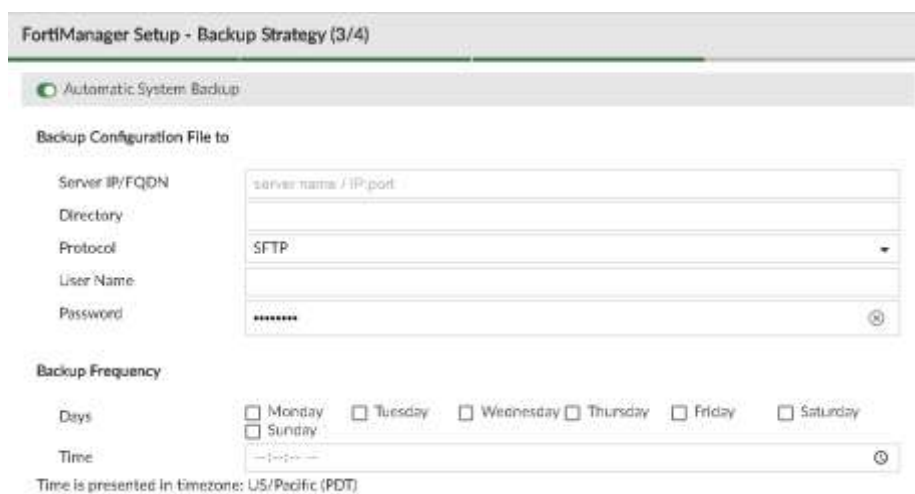


Figure 8 — The backup strategy step of the setup wizard. Configured properly in section 5 rather than here.

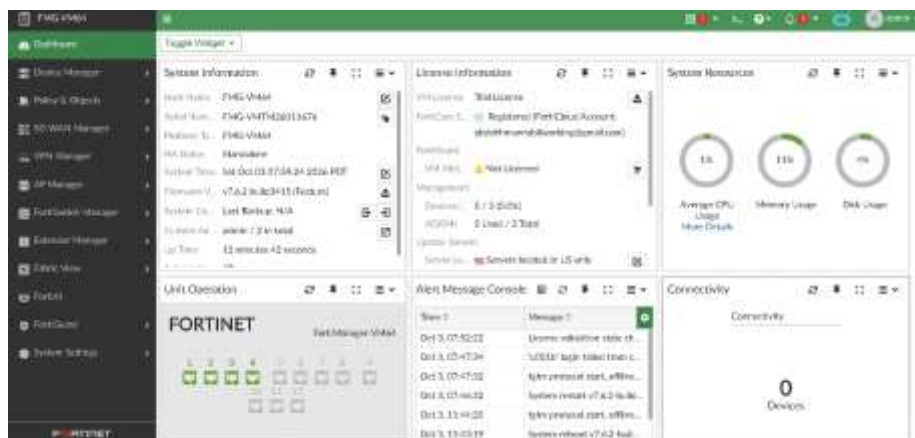


Figure 9 — The FortiManager dashboard after licensing.

03. System Settings

These settings are usually treated as housekeeping. On a management appliance they are not: FortiManager timestamps every configuration revision, every install, and every scheduled job, and those timestamps are the audit trail for changes made across the entire managed estate.

► 3.1 Hostname

The hostname appears in the GUI banner, in the CLI prompt, and inside backup filenames. On a network with more than one FortiManager — a production pair, or a lab alongside a live system — it is the only thing distinguishing one appliance's backup file from another's.

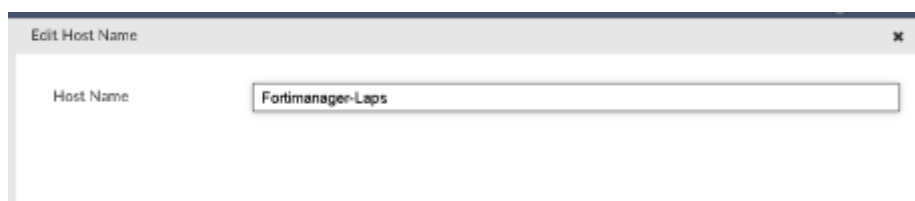


Figure 10 — Setting the hostname.

```
config system global
    set hostname "Fortimanager-Laps"
end
```

► 3.2 Time and timezone

This appliance defaulted to US/Pacific (PDT). The backup scheduler states its timezone on screen for exactly this reason: a job scheduled for 21:00 runs at 21:00 in the appliance's timezone, not the administrator's.

A wrong timezone does not announce itself. It produces backups at unexpected hours and an audit trail that disagrees with every other system in the estate. Correcting it is the first configuration change worth making.

```
config system global
    set timezone 55
end

config system ntp
    set status enable
    config ntpserver
        edit 1
            set server "pool.ntp.org"
        next
    end
end
```

```
end
end
```

System Time Sat Oct 03 20:19:54 2026
+04



Figure 11 — System time after correction, showing Asia/Muscat (+04).

► 3.3 Routing

A management interface on a static address needs a default route configured explicitly — DHCP installs one automatically, which is part of why a DHCP-configured appliance appears to work until it is converted to static.

```
Fortimanager-Laps # config system route
(route)# edit 1
(1)# set device "port1"
(1)# set gateway 172.168.0.1
(1)# end
```

Figure 12 — Configuring the default route.

```
config system route
  edit 1
    set device "port1"
    set gateway 172.168.0.1
  next
end

get system route
```

► 3.4 Administrative access settings

Separate from what the interface permits, these govern the admin session itself.

HTTP Port	80
HTTPS Port	443
HTTPS & Web Service Certificate	Fortinet_Local
Idle Timeout	60 Seconds (60-28800)
Idle Timeout (API)	900 Seconds (1-28800)
Idle Timeout (GUI)	60 Seconds (60-28800)
Remote GUI Access	☒
Access Remote GUI via Port	8082

Figure 13 — Administrative access settings — ports, certificate, and idle timeouts.

Setting	Value here	Why it matters
HTTP / HTTPS port	80 / 443	Changing these is a mild obscurity measure; it also breaks every bookmark and script that assumes the default.
Redirects to HTTPS	Disabled	Should be enabled wherever HTTP remains open, so that an accidental cleartext session is upgraded rather than served.
HTTPS certificate	Fortinet_Local	The self-signed default. Production should present a certificate the organisation's CA issued, or every admin learns to click through warnings.
Idle timeout (GUI)	60 seconds	Short. An unattended FortiManager session is authority over every managed firewall, so a short timeout is the correct default.
Idle timeout (API)	900 seconds	Longer by design — API sessions are expected to be driven by scripts rather than people.



Figure 14 — The System Settings menu, with the theme selection used to distinguish this appliance visually.

04. Administrators and Profiles

FortiManager separates the account from its permissions. An administrator is a login; an admin profile is the set of permissions attached to it. Changing what someone can do means editing a profile, not an account — which is why profiles are created first.

► 4.1 The three profile types

Type	Scope
System Admin	Permissions are granted feature by feature across the whole appliance. The most granular option, and the one used here.
Restricted Admin	A narrowed role intended for specific operational tasks rather than general administration.
ADOM Scoped Admin	Permissions apply only within assigned ADOMs. The normal choice in a multi-tenant or multi-site deployment, where an administrator should see only their own estate.

Each permission is set to **Read-Write**, **Read-Only** or **None**. Permissions set to None do not appear greyed out in the GUI — the menu entry is absent entirely, so a restricted administrator cannot see that the capability exists.

► 4.2 Building a restricted profile

The profile created here, `Lab-user`, grants read-only visibility of System Settings and the FortiGuard licence view, and nothing at all under Device Manager. It is deliberately close to a junior operator's access: able to confirm the appliance is healthy and licensed, unable to touch a managed device.

Type	Permissions		
	Read-Write	Read-Only	None
System Settings	☐	☒	☐
Administrative Domain	☐	☐	☒
FortiGuard Center	☐	☒	☐
License Management	☐	☒	☐
Firmware Management	☐	☐	☒
Settings	☐	☐	☒
Device Manager	☐	☐	☒
Add/Delete/Edit Devices/Groups	☐	☐	☒
Retrieve Configuration from Devices	☐	☐	☒
Revert Configuration from Revision History	☐	☐	☒
Delete Device Revision	☐	☐	☒
Terminal Access	☐	☐	☒
Manage Device Configurations	☐	☐	☒

Figure 15 — The Lab-user profile. System Settings and licence view are read-only; all Device Manager permissions are set to None.

► 4.3 Creating the administrator

User Name: Test-User

Avatar: [Add Photo] [Remove Photo]

Description: [Empty text box]

Admin Type: LOCAL

New Password: [Masked]

Confirm Password: [Masked]

FortiToken Cloud: [Disable] [FortiToken Mobile] [Email] [SMS]

Administrative Domain: All ADOMs | All ADOMs except specified ones | Specify

Admin Profile: Lab-user

Policy Package: All Packages | Specify

Policy Block: All Policy Blocks | Specify

JSON API Access: None

Theme Mode: Use Global Theme | Use Own Theme

High Contrast Theme: [Off]

Other Themes: [Madness] [Jade] [Neutrino] [Dark Matter] [Spring] [Summer]

Figure 16 — Creating the Test-User account and attaching the Lab-user profile.

Password strength is advisory, not enforced

The GUI warns when a password is weak but still accepts it. Enforcement comes from a password policy configured separately, under `config system password-policy` — without it, the warning is the only barrier.

► 4.4 Verifying the restriction

A profile that has been created is not a profile that has been proven. The only verification that counts is logging in as the account and comparing what it can see.

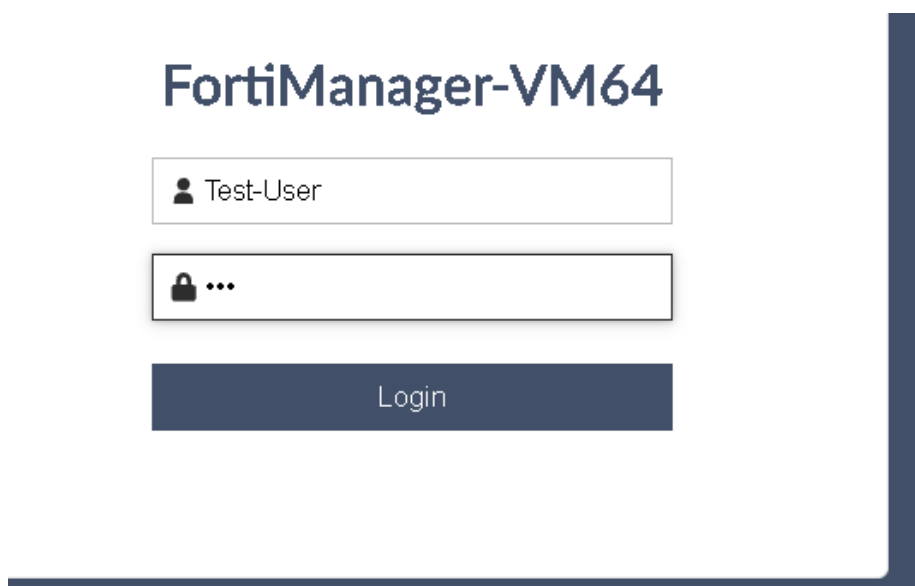


Figure 17 — Logging in as Test-User.

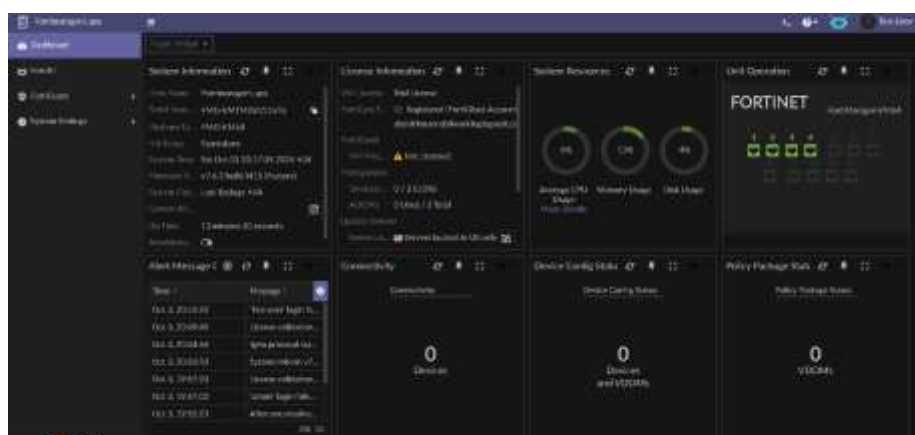


Figure 18 — The Test-User session. The navigation menu holds only Dashboard, FortiAI, FortiGuard, and System Settings.

Device Manager, Policy & Objects, SD-WAN Manager, VPN Manager and every other management module are absent from the menu entirely. Set against the administrator view

in Figure 7, which carries the full menu, this is a controlled demonstration that the profile applied — not merely that it was saved.

05. Backup and Restore

A FortiManager backup is not a device backup. It contains the device database for every managed FortiGate, every policy package, every ADOM, the revision history, and the logs. Losing it means losing the management state of the entire estate, not the configuration of one appliance.

► 5.1 Manual backup from the GUI

The System Information dashboard widget carries two arrow icons: the left one backs up, the right one restores. The widget also reports **Last Backup**, which is the quickest way to confirm that a scheduled job is actually running.

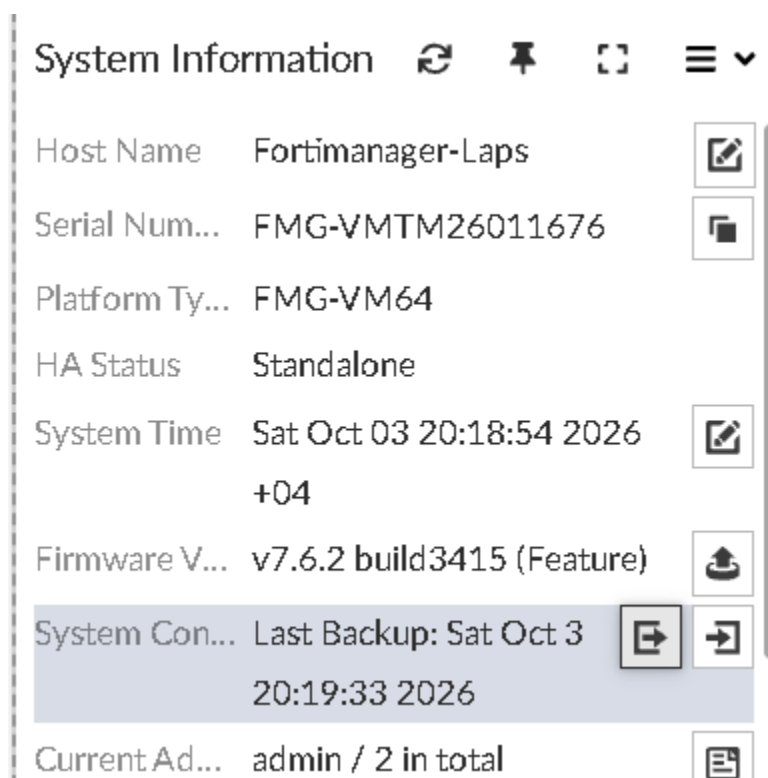


Figure 19 — The System Information widget showing the backup and restore controls and the last backup time.



Figure 20 — The Backup Now dialogue, requiring an encryption password.

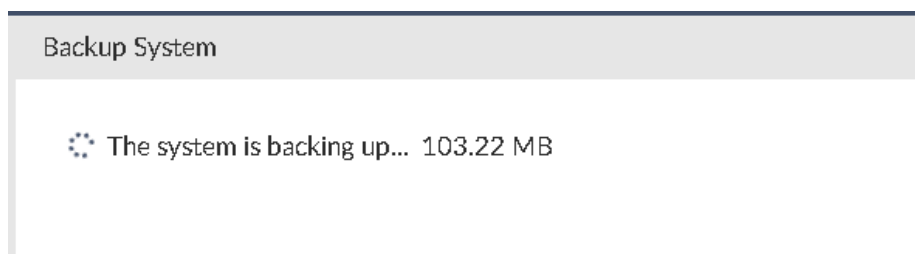


Figure 21 — The backup in progress.

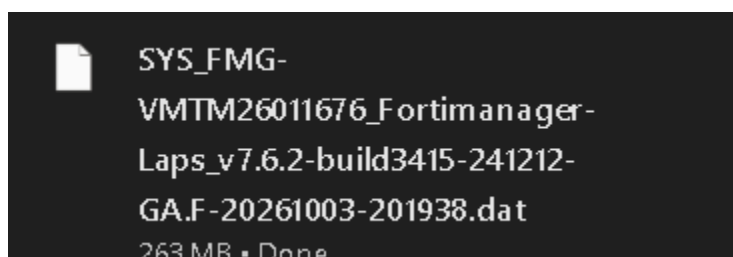


Figure 22 — The resulting file — 263 MB, with a .dat extension.

The filename encodes the serial number, hostname, firmware build and timestamp: `SYS_FMG-VMTM26011676_Fortimanager-Laps_v7.6.2-build3415-241212-GA.F-20261003-201938.dat`. On a restore that naming is what identifies which appliance and which firmware a file belongs to, because a FortiManager backup cannot be restored onto a different firmware version.

► 5.2 Hosting an SFTP target

Scheduled backups require a server to push to. The first approach — installing Windows OpenSSH Server — failed because the isolated network could not reach Windows Update to download the optional feature. A portable SFTP server was used instead.

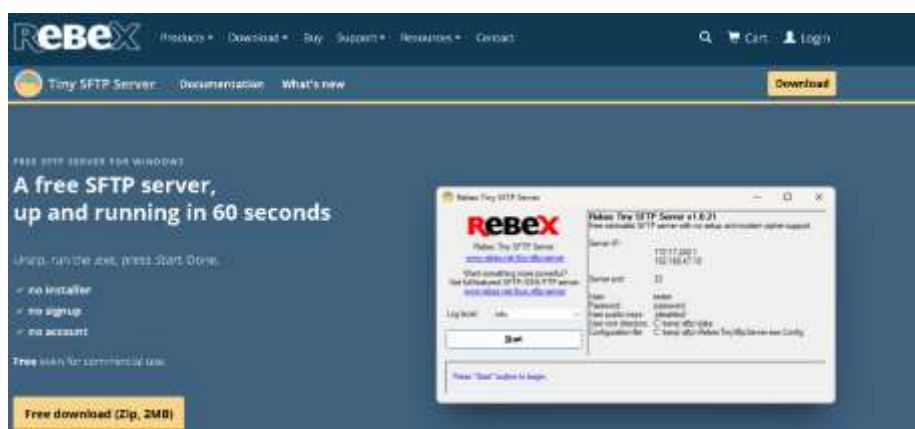


Figure 23 — Rebex Tiny SFTP Server — a portable server requiring no installation.



Figure 24 — The server on first run, generating its host keys and reporting its address, port, credentials, and root directory.

Two properties that matter for the lab

The server stores its password in cleartext in its configuration file and displays it on the main form. That is acceptable for a lab target and unacceptable for a real backup destination, which holds the management state of every firewall in the estate.

It runs as a GUI application, not a Windows service, so it only accepts connections while its window is open. A scheduled backup firing while the application is closed fails with nothing on the FortiManager side to explain why.

Connectivity was proven locally before involving FortiManager, which separates a server problem from an appliance problem.

```
PS C:\WINDOWS\system32> sftp tester@127.0.0.1
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
RSA key fingerprint is SHA256:WwaabX7OU01GniHGzDULou5WD+bmGuic1+TRDwuaEW0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

Figure 25 — First connection, accepting the server's host key.

```
(tester@127.0.0.1) Password:
Connected to 127.0.0.1.
sftp> _
```

Figure 26 — Authenticated session confirming the server accepts logins.

► 5.3 Manual backup from the CLI

The CLI form takes six arguments, and the first attempt failed on five.

Argument	Note
<server:port>	The port is part of the argument, not implied. 172.16.28.147:22, not 172.16.28.147.
<filepath>	Relative to the SFTP server's own root. Here the server's root is its data folder, so the path is simply /.
<crptpasswd>	Not optional. FortiManager encrypts the backup file, and omitting this argument is what produced return code -56.

```
execute backup all-settings sftp 172.16.28.147:22 / tester password MyEncPass123
```

```
Fortimanager-Laps # execute backup all-settings sftp 172.16.28.147:22 / tester password MyEncPass123
Starting backup all settings in background. Please wait.
Fortimanager-Laps # MD5: d6ba66f19ddfe707147d341f959c9845
Starting transfer the backup file to SFTP server...
Transferred 263.322M of 263.322M in 0:00:05s (46.347M/s)
Backup all Settings...Ok
```

Figure 27 — The corrected command: 263 MB transferred successfully.

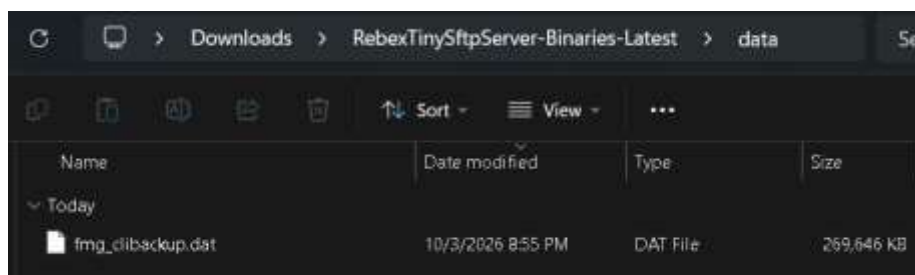


Figure 28 — The backup file landing in the SFTP server's data folder.

An encrypted backup you cannot decrypt is not a backup

The encryption password is required to restore. It is not recoverable from the appliance and it is not stored in the backup. **It must be held somewhere other than the FortiManager it protects** — a password manager or a sealed credential store — because the scenario in which the backup is needed is precisely the scenario in which the appliance is unavailable.

► 5.4 Scheduled backup

Figure 29 — The scheduled backup configuration — SFTP target, all seven days, and the timezone stated on screen.

The stored configuration can be read back from the CLI, which is the reliable way to confirm the form committed rather than merely displayed.

```

FortiManager-Laps # show system backup all-settings
config system backup all-settings
  set status enable
  set server "172.16.28.147"
  set user "tester"
  set directory "/"
  set week_days sunday monday tuesday wednesday thursday friday saturday
  set time "20:28:00"
  set passwd ENC MjQzMjM5Njc4OTMzNzg4MHLqAPYU3BgR4Jrnjx4SmKhQ/yUXnEdzUP1aLz7I1
  9t8Yax8/zgsX42Nzd2qsuY4ifY453j0501q8816HhHl.IEdLbw08QkwwQL4o6oJizLy6beRdNEzoDZ7c
  RwidpHr8Ih2Yi40Y+naMf0BEHh3PykK1/qch8Ap1BXCcxgEc
  set cryptpasswd ENC MTc1MzI5MjM0TUzNjEwMYCkHDPX80w0nd+0q3JhJYFs4oTizncgsW2BY
  6b9x8JZ7Loc1ch14J0dbF1X4dQ1x9nUpvK1jX3jJF5g4P9X+27G+D247C55a60R0C4kdkhHdqPU0HsJH
  01k7Jed8Hqk681G7N11wAcwgSyb61wUFEWtQoU99svftrBuSFgGSne
end
  
```

Figure 30 — show system backup all-settings, confirming status, target, schedule, and both encrypted passwords.

Date & Time	Admin	Size	Status	
Sat Oct 03 21:00:00 2026	Scheduled	263.3 MB	Success	

Figure 31 — Backup History confirming the scheduled job ran and completed — 263.3 MB, status Success.

► 5.5 Findings

Symptom	A scheduled backup set for a time three minutes ahead did not run. <code>show system backup all-settings</code> confirmed the schedule was stored and enabled; Backup History stayed empty and nothing arrived at the SFTP server.
Cause	FortiManager computes the next run when the configuration is committed, and its scheduler evaluates on an interval rather than continuously. A target time only minutes ahead can fall outside the next evaluation, deferring the job to the following day.
Fix	The same schedule set with a longer margin ran on time and completed successfully.
Rule	A scheduled job that has not yet fired proves nothing. Verification means a populated Backup History entry and a file present at the destination — not a saved form.

A second observation is operational rather than diagnostic. This backup is 263 MB on an appliance managing **zero** devices, scheduled seven days a week — approximately 1.8 GB per week before a single FortiGate is registered, and it grows with every device, policy package and revision added. FortiManager does not prune what it sends, so retention is the destination server's responsibility.

o6. Offline Mode

Offline mode stops FortiManager initiating outbound connections to FortiGuard and FortiCare. The appliance continues to run and continues to manage registered devices; what ends is its own outbound chatter.

Use case	Why offline mode fits
Air-gapped or segmented networks	An appliance with no path to the internet otherwise generates continuous failed connection attempts and alert noise.
Staging before deployment	A FortiManager configured in advance and shipped to site should not attempt to register or update until it reaches its final network.
OT and industrial environments	Outbound traffic from a management appliance may be prohibited by the network's security architecture rather than merely unavailable.
Licence-validation noise	Repeated validation failures on a disconnected appliance fill the event log and obscure real events.

Offline mode must be disabled before devices are registered in later labs, since the FGFM management tunnel and FortiGuard-dependent features are affected by it.

Workspace	Mail Server	Syslog Server	Meta Fields	Misc Settings	
Advanced Settings					

Offline Mode ⓘ



And you can download the license and upload it to fortimanager and also get rid of the license noise

FMG-VMTM26011676.lic	10/3/2026 8:03 PM	LIC File
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Ports

Port	Protocol	Purpose
TCP 541	FGFM	The FortiGate-to-FortiManager management tunnel. The single most important port in the product.
TCP 542	FGFM (IPv6)	The same tunnel over IPv6.
TCP 443	HTTPS	Administrative GUI, and the JSON-RPC API.
TCP 22	SSH	Administrative CLI.
TCP 8080 / 443	FortiGuard	Outbound to FortiGuard, and inbound from managed devices when FortiManager acts as a local FDS.

o8. Summary

Area	Outcome	Evidence
Management interface	Configured and reachable	Console configuration and get system interface output
Licensing	Activated	Trial licence registered to FortiCare; 3 devices and 2 ADOMs available
Hostname, timezone, NTP	Corrected from US/Pacific to Asia/Muscat	Dashboard system time at +04
Default route	Configured	CLI configuration
Admin access settings	Reviewed and documented	Ports, certificate, and idle timeouts
Admin profile and account	Verified by login	Test-User session showing a reduced navigation menu against the administrator view
Manual backup — GUI	Succeeded	263 MB .dat file; dashboard Last Backup populated
Manual backup — CLI	Succeeded	263.325 MB transferred to SFTP after correcting the argument list
Scheduled backup	Succeeded	Backup History entry at 21:00, 263.3 MB, status Success
Scheduler timing	Documented finding	A schedule set three minutes ahead did not fire; the same schedule with margin did
Offline mode	Documented	Command and deployment scenarios recorded